

EXPOSÉ IV

LIFTING K3 SURFACES TO CHARACTERISTIC ZERO

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Rudakov and Shafarevitch [14] (see also Nygaard [12]) established that K3 surfaces admit no nonzero vector fields.

The purpose of this exposé is to prove, from this, that every polarized K3 surface in characteristic $p > 0$ lifts to characteristic zero (1.8). The proof rests on the fact that the Rudakov–Shafarevitch result entails, for K3 surfaces, the existence of a good deformation theory, making it possible in particular to “control” the locus in the versal formal space along which a given nontrivial invertible sheaf deforms. The only ingredients used are, on the one hand, the usual relations in de Rham cohomology between the Gauss–Manin connection and the Kodaira–Spencer operation, and, on the other hand, the existence of a crystalline Chern class for invertible sheaves.

We denote by k an algebraically closed field of characteristic $p > 0$, and write $W = W(k)$ for the ring of Witt vectors over k .

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1. STATEMENT OF THE LIFTING THEOREM

In this section, X_0 denotes a K3 surface over k .

PROPOSITION 1.1.

a) The Hodge spectral sequence

$$E_1^{ij} = H^j(X_0, \Omega_{X_0/k}^i) \implies H_{\text{DR}}^*(X_0/k)$$

degenerates at E_1 , and the matrix of Hodge numbers $h^{ij} = \dim_k H^j(X_0, \Omega_{X_0/k}^i)$ is

$$\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 20 & 0 \\ 1 & 0 & 1 \end{array}.$$

b) Let $T = T_{X_0/k} = (\Omega_{X_0/k}^1)^\vee$ be the tangent bundle of X_0 . Then $H^i(X_0, T) = 0$ for $i = 0$ or 2 , and $\dim_k H^1(X_0, T) = 20$.

c) The crystalline cohomology W -modules $H^i(X_0/W)$ are free, of ranks $1, 0, 22, 0, 1$ for $i = 0, 1, 2, 3, 4$.

By [14], $H^0(X_0, T) = 0$. By the definition of a K3 surface, $\Omega_{X_0/k}^2$ is trivial; hence

$$T_{X_0/k} \simeq \Omega_{X_0/k}^1, \tag{1.1.1}$$

and consequently $H^0(X_0, \Omega_{X_0/k}^1) = 0$, so that $H^2(X_0, \Omega_{X_0/k}^1) = 0$ by Serre duality. Moreover, $H^1(X_0, \mathcal{O}) = 0$ and $H^2(X_0, \mathcal{O}) \simeq k$ by the definition of a K3 surface. Thus the table of the Hodge numbers h^{ij} , for $(i, j) \neq (1, 1)$, is the one displayed in 1.1(a). It follows that the Hodge spectral sequence degenerates at E_1 . Recall also (SGA 5, VII 4.11) that

$$\sum (-1)^{i+j} h^{ij} = c_2,$$

and that $c_2 = 24$ by Riemann–Roch. Hence $h^{11} = 20$, which gives (b), in view of (1.1.1). It follows from (a) that the spaces $H_{\text{DR}}^i(X_0/k)$ have dimensions

1, 0, 22, 0, 1 for $i = 0, 1, 2, 3, 4$. Assertion (c) follows from the “universal-coefficients” exact sequence [2, VII 1.1.11]:

$$0 \longrightarrow H^i(X_0/W) \otimes k \longrightarrow H_{\text{DR}}^i(X_0/k) \longrightarrow \text{Tor}_1^W(H^{i+1}(X_0/W), k) \longrightarrow 0.$$

COROLLARY 1.2. The versal formal space S of deformations of X_0 over local Artinian W -algebras with residue field k [15] is universal and formally smooth of dimension 20; that is, it is W -isomorphic to $\text{Spf}(W[[t_1, \dots, t_{20}]])$.

This is an immediate consequence of 1.1(b).

In what follows in this section, we write

$$\underline{X}/S \tag{1.3}$$

for the universal deformation of X_0 over S .

1.4. Let L_0 be an invertible sheaf on X_0 . Write $\text{Def}(X_0, L_0)$ for the functor on the category $\mathcal{A}$ of local Artinian W -algebras with residue field k that assigns to each $A \in \mathcal{A}$ the set of isomorphism classes of pairs (X, L) consisting of a flat (hence smooth) deformation X of X_0 over A and an extension of L_0 to an invertible sheaf L on X . Write $\text{Def}(X_0)$ for the functor assigning to each $A \in \text{Ob } \mathcal{A}$ the set of isomorphism classes of flat deformations of X_0 over A ; by 1.2 this functor is (pro-)represented by S . There is a morphism “forget the extension of L_0 ”

$$\text{Def}(X_0, L_0) \longrightarrow \text{Def}(X_0). \tag{1.4.1}$$

PROPOSITION 1.5. The functor $\text{Def}(X_0, L_0)$ is pro-representable, and the morphism (1.4.1) is a closed immersion defined by one equation.

The first assertion means that there is a largest closed formal subscheme

$$\Sigma(L_0) \subset S \tag{1.5.0}$$

such that L_0 extends to an invertible sheaf over $\underline{X} \times_S \Sigma(L_0)$, and such that this extension is unique.

It is readily verified that the functor $\text{Def}(X_0, L_0)$ satisfies Schlessinger's conditions for the existence of a hull. Thus there is a formal scheme $S' = \text{Spf}(R')$, where R' is a complete Noetherian local W -algebra with residue field k , and a deformation $(\underline{X}', \underline{L}')$ of (X_0, L_0) over S' such that, for every $A \in \text{Ob } \mathcal{A}$, the associated map

$$\text{Hom}(R', A) \longrightarrow \text{Def}(X_0, L_0)(A) \quad (1.5.1)$$

is surjective, and is bijective for $A = k[\varepsilon]$, $\varepsilon^2 = 0$. Let R be the ring of S . Since S pro-represents $\text{Def}(X_0)$, $\underline{X}'$ defines a homomorphism $u : R \rightarrow R'$ such that the composite

$$\text{Hom}(R', A) \xrightarrow{1.5.1} \text{Def}(X_0, L_0)(A) \xrightarrow{1.4.1} \text{Def}(X_0)(A) = \text{Hom}(R, A) \quad (1.5.2)$$

is $\text{Hom}(u, A)$. To establish the first assertion of 1.5, it is therefore enough to prove that u is surjective, since then (1.5.2) is injective and hence (1.5.1) is bijective. By a familiar lemma [15, 1.1], this is equivalent to proving that, if $\mathfrak{m}$ (respectively $\mathfrak{m}'$) is the maximal ideal of R (respectively R'), the map $\mathfrak{m}/(pR + \mathfrak{m}^2) \rightarrow \mathfrak{m}'/(pR' + \mathfrak{m}'^2)$ induced by u is surjective; equivalently, that the tangent map at the origin of (1.4.1),

$$\text{Hom}(u, k[\varepsilon]) : \text{Def}(X_0, L_0)(k[\varepsilon]) \longrightarrow \text{Def}(X_0)(k[\varepsilon]), \quad (1.5.3)$$

is injective. Let T' be the sheaf on X_0 of automorphisms of the trivial deformation of (X_0, L_0) over $k[\varepsilon]$. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_{X_0} \longrightarrow T' \longrightarrow T \longrightarrow 0, \quad (1.5.4)$$

where $T' \rightarrow T$ is defined by forgetting L_0 , and $T = T_{X_0}$ is viewed as the sheaf of automorphisms of the trivial deformation of X_0 over $k[\varepsilon]$. It is standard that (1.5.3) identifies canonically with the map

$H^1(X_0, T') \rightarrow H^1(X_0, T)$ induced by $T' \rightarrow T$. Since $H^1(X_0, \mathcal{O}) = 0$, the long exact cohomology sequence of (1.5.4) shows that (1.5.3) is injective. It remains to prove that the closed immersion $S' \hookrightarrow S$ is defined by a single equation, i.e. that $I = \ker(u)$ is principal. Put $S'' = \mathrm{Spf}(R/\mathfrak{m}I)$; this is a square-zero thickening of S' in S , with ideal $I/\mathfrak{m}I$. The obstruction to extending the invertible sheaf $\underline{L}'$ above $\underline{X} \times_S S''$ is an element

$$a \in H^2(X_0, I/\mathfrak{m}I) = H^2(X_0, \mathcal{O}) \otimes I/\mathfrak{m}I,$$

which, after choosing a basis of $H^2(X_0, \mathcal{O})$, we regard as an element of $I/\mathfrak{m}I$. Let $\Sigma = \mathrm{Spf}(R/(\mathfrak{m}I + (f)))$, where $f \in I$ lifts a . Then $S' \subset \Sigma \subset S'' \subset S$, and by construction (and functoriality of the obstruction) $\underline{L}'$ extends to $\underline{X} \times_S \Sigma$. The universal property of S' therefore implies $S' = \Sigma$, i.e. $\mathfrak{m}I + (f) = I$. Nakayama's lemma now shows that f generates I , completing the proof.

We may now state the main result of the exposé.

THEOREM 1.6. Let L_0 be a nontrivial invertible sheaf on X_0 . With the notation of 1.5, the formal scheme $\Sigma(L_0)$ pro-representing $\mathrm{Def}(X_0, L_0)$ is flat over W , of relative dimension 19.

In other words, if f is an equation for $\Sigma(L_0)$ in S (1.5), then p does not divide f . Equivalently, $\Sigma(L_0)$ does not contain the reduction S_0 of S modulo p , i.e. L_0 cannot extend over $\underline{X} \times_S S_0$.

The proof of 1.6 will be given in Section 2. We end this section with a few consequences of 1.6.

COROLLARY 1.7. Let L_0 be a nontrivial invertible sheaf on X_0 . There exist a trait T finite over W , a deformation of X_0 to a formal scheme $\underline{X}$ flat over T , and an extension of L_0 to an invertible sheaf $\underline{L}$ on $\underline{X}$.

We must prove that there is a W -morphism $T \rightarrow \Sigma(L_0)$, where T is a trait finite over W . Since p is not a zero divisor in the ring R' of $\Sigma(L_0)$, there exist (EGA 0_{IV} 16.4.1) elements $x_1, \dots, x_n$ of the maximal ideal of R' which, together with p , form a system of parameters for R' (thus $n = 19$). The quotient $B = R'/(x_1, \dots, x_n)$ is quasi-finite over W , hence finite over W . There is therefore a local W -homomorphism from B to a complete discrete valuation ring C finite over W , and the composite $R' \rightarrow B \rightarrow C$ gives the required morphism.

Applying Grothendieck's algebraization theorem (EGA III 5.4.5), one deduces from 1.7 the announced lifting theorem.

COROLLARY 1.8. Let L_0 be an ample invertible sheaf on X_0 . There exist a trait T finite over W , a deformation of X_0 to a proper smooth scheme X over T , and an extension of L_0 to an ample invertible sheaf L on X .

REMARK 1.9. It is not known whether every K3 surface over k lifts to a proper smooth scheme over W . Ogus [13, §2] shows that: (a) every K3 surface over k lifts over W , except possibly in the “superspecial”, non-elliptic case, which in fact does not exist if one assumes Artin's conjecture [1]; (b) if $p > 2$, every K3 surface over k lifts over $W[\sqrt{p}]$. Thus, at present, only the special case of 1.8 in which $p = 2$ and X_0 is superspecial is not covered by these results. We also point out that the cited paper of Ogus contains useful further information on the structure of $\Sigma(L_0)$; see also the following exposé for the case in which X_0 is ordinary.

COROLLARY 1.10. If k is the algebraic closure of a finite field over which X_0 is defined, the corresponding Frobenius acts semisimply on $H^2(X_0, \mathbf{Q}_\ell)$, where $\ell \neq p$.

This follows from [5]. With the notation of 1.8, the generic fibre of X is a K3 surface (the property of a surface of being K3 is stable under generization, since it is expressed by “ K algebraically equivalent to zero and $\chi(\mathcal{O}) = 2$ ”). Hence X_0 satisfies the hypothesis of [5, 1.2]; the conclusion follows from [5, 6.6] and from the fact that Frobenius acts semisimply on the ℓ -adic H^1 of an abelian variety over a finite field ([16], [11, p. 203]). Notice that 1.10 is in fact independent of the assertion $H^0(X_0, T_{X_0}) = 0$: if X_0 admitted a nonzero vector field, then X_0 would be unirational by [14], and the conclusion of 1.10 would still hold (trace argument).

2. DE RHAM COHOMOLOGY OF $\underline{X}/S$ AND PROOF OF 1.6

We retain the notation of Section 1: X_0 is a K3 surface over k , and $\underline{X}/S$ denotes its universal W -deformation (1.3). Readers familiar with de Rham cohomology may skip 2.1–2.10, which merely recall standard facts concerning the Gauss–Manin connection, the Hodge filtration, the Frobenius operation, and the Chern class of an invertible sheaf.

2.1. Let $\Omega_{\underline{X}/S}^\bullet$ be the de Rham complex of the relative formal scheme $\underline{X}/S$ (by definition, for each i , $\Omega_{\underline{X}/S}^i$ is the inverse limit of the usual modules of differentials $\Omega_{\underline{X} \times_S S'/S'}^i$, as S' runs through the infinitesimal neighbourhoods of $\text{Spec}(k)$ in S). Let $f : \underline{X} \rightarrow S$ be the projection. By definition, the de Rham cohomology of $\underline{X}/S$ consists of the $\mathcal{O}_S$ -modules

$$H_{\text{DR}}^i(\underline{X}/S) \stackrel{\text{def}}{=} R^i f_*(\Omega_{\underline{X}/S}^\bullet), \quad (2.1.1)$$

whereas the Hodge cohomology of $\underline{X}/S$ consists of the $\mathcal{O}_S$ -modules

$$H^i(\underline{X}, \Omega_{\underline{X}/S}^j) \stackrel{\text{def}}{=} R^i f_*(\Omega_{\underline{X}/S}^j). \quad (2.1.2)$$

Since the $\mathcal{O}_{\underline{X}}$ -modules $\Omega_{\underline{X}/S}^i$ are locally free of finite type, the $\mathcal{O}_S$ -modules (2.1.1) and (2.1.2) are coherent by Grothendieck's finiteness theorem (EGA III 3.4.2). There is also the usual spectral sequence (the “Hodge spectral sequence”)

$$E_1^{ij} = H^j(\underline{X}, \Omega_{\underline{X}/S}^i) \Longrightarrow H_{\text{DR}}^*(\underline{X}/S). \quad (2.1.3)$$

PROPOSITION 2.2.

- a) The spectral sequence (2.1.3) degenerates at E_1 ; the $\mathcal{O}_S$ -modules $H^j(\underline{X}, \Omega_{\underline{X}/S}^i)$ are free of finite type, and the canonical maps

$$H^j(\underline{X}, \Omega_{\underline{X}/S}^i) \otimes k \longrightarrow H^j(X_0, \Omega_{X_0/k}^i)$$

are isomorphisms.

- b) The $\mathcal{O}_S$ -modules $H_{\text{DR}}^i(\underline{X}/S)$ are free of finite type, and the canonical maps

$$H_{\text{DR}}^i(\underline{X}/S) \otimes k \longrightarrow H_{\text{DR}}^i(X_0/k)$$

are isomorphisms.

- c) The cup product

$$\langle , \rangle : H_{\text{DR}}^2(\underline{X}/S) \otimes H_{\text{DR}}^2(\underline{X}/S) \longrightarrow H_{\text{DR}}^4(\underline{X}/S)$$

is a perfect duality.

Because the table of Hodge numbers of X_0/k is “interlaced with zeros” (1.1(a)), the usual criterion (EGA III 7.5.4), applied to the cohomological functors $M \mapsto H^\bullet(\underline{X}, \Omega_{\underline{X}/S}^i \otimes f^*M)$, immediately gives the second assertion of (a), and the first follows. Assertion (b) follows from (a), and implies (c) by Poincaré duality for $H_{\text{DR}}^*(X_0/k)$.

2.3. Let $\Omega_{S/W}^\bullet$ be the de Rham complex of the “formal” differentials of S/W : $\Omega_{S/W}^i = \bigwedge^i \Omega_{S/W}^1$, where $\Omega_{S/W}^1$, the inverse limit of the completed modules of differentials Ω_{S_n/W_n}^1 (with S_n/W_n the reduction of S/W modulo p^{n+1}), is free over $\mathcal{O}_S$ with basis $dt_1, \dots, dt_{20}$ if $S \simeq W[[t_1, \dots, t_{20}]]$. The $H_{\text{DR}}^i(\underline{X}/S)$ carry a canonical integrable connection, the Gauss–Manin connection,

$$\nabla : H_{\text{DR}}^i(\underline{X}/S) \longrightarrow \Omega_{S/W}^1 \otimes H_{\text{DR}}^i(\underline{X}/S). \quad (2.3.1)$$

The simplest definition of (2.3.1) paraphrases the Katz–Oda construction [], starting from the canonical extension

of $\mathcal{O}_{\underline{X}}$ -modules

$$0 \longrightarrow f^* \Omega_{S/W}^1 \longrightarrow \Omega_{\underline{X}/W}^1 \longrightarrow \Omega_{\underline{X}/S}^1 \longrightarrow 0, \quad (2.3.0)$$

where $\Omega_{\underline{X}/W}^1$ is the module of completed differentials. One may also use the fact that $H_{\text{DR}}^i(\underline{X}/S)$ is the value at S of a crystal in $\mathcal{O}$ -modules on the crystalline site of S_0/W :

$$H_{\text{DR}}^i(\underline{X}/S) = R^i(f_0)_{\text{cris}*}(\mathcal{O}_{\underline{X}_0/W})(S), \quad (2.3.2)$$

where

$$(f_0 : \underline{X}_0 \rightarrow S_0) = f \times_S S_0, \quad S_0 = S \otimes_W k \simeq \text{Spf}(k[[t_1, \dots, t_{20}]]), \quad (2.3.2.1)$$

and $(f_0)_{\text{cris}} : (\underline{X}_0/W)_{\text{cris}} \rightarrow (S_0/W)_{\text{cris}}$ is the corresponding morphism of crystalline sites. This is a “formal variant” of Berthelot’s result [2, V 3.6], readily deduced from it by applying it to the morphisms induced by f_0 over the infinitesimal neighbourhoods of $\text{Spec}(k)$ in S_0 .

Either definition shows that the cup product of 2.2(c) is horizontal:

$$\langle \nabla x, y \rangle + \langle x, \nabla y \rangle = \nabla \langle x, y \rangle \quad (2.3.3)$$

for all $x, y \in H_{\text{DR}}^2(\underline{X}/S)$.

Let $F_{\text{Hdg}}^i H_{\text{DR}}^2(\underline{X}/S)$ be the Hodge filtration, the abutment of (2.1.3). If $\Omega_{\underline{X}/S}^{\geq i} = (0 \rightarrow \Omega_{\underline{X}/S}^i \rightarrow \Omega_{\underline{X}/S}^{i+1} \rightarrow \dots)$ denotes the de Rham complex truncated by zero in degrees $< i$, then the inclusion $\Omega_{\underline{X}/S}^{\geq i} \rightarrow \Omega_{\underline{X}/S}^\bullet$ gives, by 2.2(a), an isomorphism

$$H^2(\underline{X}, \Omega_{\underline{X}/S}^{\geq i}) \simeq F_{\text{Hdg}}^i H_{\text{DR}}^2(\underline{X}/S). \quad (2.3.4)$$

We have

$$H_{\text{DR}}^2(\underline{X}/S) = F_{\text{Hdg}}^0 \supset F_{\text{Hdg}}^1 \supset F_{\text{Hdg}}^2 \supset F_{\text{Hdg}}^3 = 0, \quad (2.3.5)$$

the F_{Hdg}^i are free $\mathcal{O}_S$ -modules, and degeneration of (2.1.3) gives isomorphisms

$$H^{2-i}(\underline{X}, \Omega_{\underline{X}/S}^i) \simeq \text{gr}_F^i H_{\text{DR}}^2(\underline{X}/S), \quad (F = F_{\text{Hdg}}). \quad (2.3.6)$$

In particular, F_{Hdg}^1 (respectively F_{Hdg}^2) is free of rank 21 (respectively 1). From (2.3.4) one sees that the orthogonal of F_{Hdg}^2 for the cup product of 2.2(c) contains F_{Hdg}^1 , and therefore equals it:

$$F_{\text{Hdg}}^1 = (F_{\text{Hdg}}^2)^\perp \quad (\text{and hence } F_{\text{Hdg}}^2 = (F_{\text{Hdg}}^1)^\perp). \quad (2.3.7)$$

The Katz–Oda description of the Gauss–Manin connection shows that

$$\nabla F_{\text{Hdg}}^i H_{\text{DR}}^2(\underline{X}/S) \subset \Omega_{S/W}^1 \otimes F_{\text{Hdg}}^{i-1} H_{\text{DR}}^2(\underline{X}/S) \quad (2.3.8)$$

(“Griffiths transversality”; see, for example, [8, 1.4]). Passing to quotients, ∇ therefore induces an $\mathcal{O}_S$ -linear map

$$\text{gr}^i \nabla : \text{gr}_F^i H_{\text{DR}}^2(\underline{X}/S) \longrightarrow \Omega_{S/W}^1 \otimes \text{gr}_F^{i-1} H_{\text{DR}}^2(\underline{X}/S). \quad (2.3.9)$$

Correspondingly there is an $\mathcal{O}_S$ -linear map

$$\nabla_i : T_{S/W} \longrightarrow \text{Hom}(\text{gr}_F^i H_{\text{DR}}^2(\underline{X}/S), \text{gr}_F^{i-1} H_{\text{DR}}^2(\underline{X}/S)), \quad (2.3.10)$$

where $T_{S/W} = (\Omega_{S/W}^1)^\vee$. By (2.3.6), the target of (2.3.10) identifies canonically, for the fixed i , with

$$\text{Hom}(H^{2-i}(\underline{X}, \Omega_{\underline{X}/S}^i), H^{3-i}(\underline{X}, \Omega_{\underline{X}/S}^{i-1})).$$

Let $T_{\underline{X}/S} = (\Omega_{\underline{X}/S}^1)^\vee$. If

$$\text{Kod}(\underline{X}/S) : T_{S/W} \longrightarrow H^1(\underline{X}, T_{\underline{X}/S}) \quad (2.3.11)$$

denotes the Kodaira–Spencer map associated with $\underline{X}/S/W$, defined by the extension (2.3.0) (cf. [8, 1.3]), then, under the preceding identification, (2.3.10) belongs to the commutative triangle

$$\begin{array}{ccc} T_{S/W} & \xrightarrow{\text{Kod}(\underline{X}/S)} & H^1(\underline{X}, T_{\underline{X}/S}) \\ & \searrow \nabla_i & \downarrow \\ & & \text{Hom}(H^{2-i}(\underline{X}, \Omega_{\underline{X}/S}^i), H^{3-i}(\underline{X}, \Omega_{\underline{X}/S}^{i-1})), \end{array} \quad (2.3.12)$$

where the vertical arrow is defined by cup product, via the interior product $T_{\underline{X}/S} \otimes \Omega_{\underline{X}/S}^i \rightarrow \Omega_{\underline{X}/S}^{i-1}$. This compatibility is verified exactly as

in [8, 1.4.1.7].

By (2.3.7), the cup product of 2.2(c), after passage to gr_F , induces a perfect duality

$$\mathrm{gr}_F^i H_{\mathrm{DR}}^2(\underline{X}/S) \otimes \mathrm{gr}_F^{2-i} H_{\mathrm{DR}}^2(\underline{X}/S) \longrightarrow H_{\mathrm{DR}}^4(\underline{X}/S) \simeq H^2(\underline{X}, \Omega_{\underline{X}/S}^2), \quad (2.3.13)$$

corresponding under (2.3.6) to the cup product

$$H^{2-i}(\underline{X}, \Omega_{\underline{X}/S}^i) \otimes H^i(\underline{X}, \Omega_{\underline{X}/S}^{2-i}) \longrightarrow H^2(\underline{X}, \Omega_{\underline{X}/S}^2).$$

It follows from (2.3.3) that, for every $D \in T_{S/W}$,

$$\nabla_1(D) = -\nabla_2(D)^\vee, \quad (2.3.14)$$

i.e. $\langle \nabla_2(D)x, y \rangle + \langle x, \nabla_1(D)y \rangle = 0$ for $x \in \mathrm{gr}_F^2$, $y \in \mathrm{gr}_F^1$.

The following statement expresses the mobility of the Hodge filtration under Gauss–Manin.

PROPOSITION 2.4. The maps (2.3.10) and (2.3.11) are isomorphisms.

First, the sheaf $\Omega_{\underline{X}/S}^2$, the extension to $\underline{X}$ of the trivial sheaf $\Omega_{X_0/k}^2$, is necessarily trivial (1.5); hence

$$T_{\underline{X}/S} \simeq \Omega_{\underline{X}/S}^1. \quad (2.4.1)$$

Consequently, by 2.2(a), $H^1(\underline{X}, T_{\underline{X}/S})$ is free of rank 20, and the canonical map $H^1(\underline{X}, T_{\underline{X}/S}) \otimes k \rightarrow H^1(X_0, T_{X_0/k})$ is an isomorphism. A standard argument shows that

$$\mathrm{Kod}(\underline{X}/S) \otimes_W k : T_{S/W} \otimes_W k \longrightarrow H^1(X_0, T_{X_0/k}) \quad (2.4.2)$$

is precisely the tangent map at the origin of the canonical isomorphism $S \simeq \mathrm{Def}(X_0)$ (1.4), so $\mathrm{Kod}(\underline{X}/S)$ is an isomorphism. By (2.3.12), it remains to prove that the vertical arrow in (2.3.12) is an isomorphism; by (2.3.14), it is enough to do so for $i = 1$. By 2.2(a) and the observation at the beginning of the proof, it suffices to show that the cup product

$$H^1(X_0, T_{X_0/k}) \otimes H^1(X_0, \Omega_{X_0/k}^1) \longrightarrow H^2(X_0, \mathcal{O}) \quad (2.4.3)$$

is nondegenerate. But the choice of a basis of $H^0(X_0, \Omega_{X_0/k}^2)$ identifies $T_{X_0/k}$ with $\Omega_{X_0/k}^1$ and identifies (2.4.3) with Serre duality

$$H^1(X_0, \Omega_{X_0/k}^1) \otimes H^1(X_0, \Omega_{X_0/k}^1) \longrightarrow H^2(X_0, \Omega_{X_0/k}^2),$$

which proves the assertion.

COROLLARY 2.4.4. The map $\mathrm{gr}^1 \nabla$ (respectively $\mathrm{gr}^2 \nabla$) in (2.3.9) is an isomorphism (respectively is injective with free cokernel).

Consequently the same assertion holds for $\mathrm{gr}^1(\nabla|S_0)$ (respectively $\mathrm{gr}^2(\nabla|S_0)$), where $\nabla|S_0$ is the Gauss–Manin connection on $H_{\mathrm{DR}}^2(\underline{X}_0/S_0)$. In view of the duality (2.3.13), 2.4.4 follows immediately from the fact that (2.3.10) is an isomorphism.

2.5. With the notation of (2.3.2.1), let $F_{\underline{X}_0/S_0} : \underline{X}_0 \rightarrow \underline{X}_0^{(p)}$ be the relative Frobenius of $\underline{X}_0/S_0$, defined by the usual Cartesian diagram

$$\begin{array}{ccccc} \underline{X}_0 & \longleftarrow & \underline{X}_0^{(p)} & \xleftarrow{F_{\underline{X}_0/S_0}} & \underline{X}_0 \\ \downarrow & & \downarrow & \swarrow & \\ S_0 & \longleftarrow & S_0 & & \end{array} \quad (2.5.1)$$

where the horizontal composites are the absolute Frobenius maps. The crystalline interpretation (2.3.2) of $H_{\mathrm{DR}}^i(\underline{X}/S)$ shows, by functoriality of crystalline cohomology, that $F_{\underline{X}_0/S_0}$ induces, for every lift $\varphi : S \rightarrow S$ of the absolute Frobenius of S_0 compatible with the canonical Frobenius σ of W , an $\mathcal{O}_S$ -linear horizontal homomorphism

$$F(\varphi) : \varphi^* H_{\mathrm{DR}}^i(\underline{X}/S) \longrightarrow H_{\mathrm{DR}}^i(\underline{X}/S). \quad (2.5.2)$$

This homomorphism is an isogeny (i.e. $F(\varphi) \otimes_{\mathbf{Z}_p} \mathbf{Q}_p$ is an isomorphism), and its dependence on φ is controlled by ∇ (cf. [7] and the following exposé). In what follows we fix a lift φ , for example the one given by $\varphi(t_i) = t_i^p$ ($1 \leq i \leq 20$), with $S = \text{Spf } W[[t_1, \dots, t_{20}]]$, so that $\varphi(\sum a_\alpha t^\alpha) = \sum a_\alpha^\sigma t^{p\alpha}$. We write $F(\varphi) = \tilde{F}$ and denote by

$$F : H_{\text{DR}}^i(\underline{X}/S) \longrightarrow H_{\text{DR}}^i(\underline{X}/S) \quad (2.5.3)$$

the σ -linear homomorphism obtained by composing $\tilde{F}$ with the adjunction map $H_{\text{DR}}^i(\underline{X}/S) \rightarrow \varphi^* H_{\text{DR}}^i(\underline{X}/S)$, $x \mapsto 1 \otimes x$. By construction, F is compatible with the cup product of 2.2(c):

$$\langle Fx, Fy \rangle = F\langle x, y \rangle \quad (2.5.4)$$

for all $x, y \in H_{\text{DR}}^2(\underline{X}/S)$.

The following statement is a special case of a theorem of Mazur–Ogus ([10], [4, 8.26])^{*}.

PROPOSITION 2.6. With the notation of 2.3 and 2.5,

$$F_{\text{Hdg}}^1 H_{\text{DR}}^2(\underline{X}/S) \subset \{x \in H_{\text{DR}}^2(\underline{X}/S) \mid Fx \in p H_{\text{DR}}^2(\underline{X}/S)\},$$

and the two sides have the same image in $H_{\text{DR}}^2(\underline{X}_0/S_0)$.

In other words, if $F : H_{\text{DR}}^2(\underline{X}_0/S_0) \rightarrow H_{\text{DR}}^2(\underline{X}_0/S_0)$ denotes the p -linear Frobenius endomorphism (the reduction modulo p of (2.5.3)), and if $F_{\text{Hdg}}^i H_{\text{DR}}^2(\underline{X}_0/S_0)$ is the Hodge filtration (the reduction modulo p of that on $H_{\text{DR}}^2(\underline{X}/S)$), then

$$F_{\text{Hdg}}^1 H_{\text{DR}}^2(\underline{X}_0/S_0) = \ker(F : H_{\text{DR}}^2(\underline{X}_0/S_0) \rightarrow H_{\text{DR}}^2(\underline{X}_0/S_0)). \quad (2.6.1)$$

We recall the proof. By definition, F is the composite of the canonical p -linear injection $H_{\text{DR}}^2(\underline{X}_0/S_0) \rightarrow H_{\text{DR}}^2(\underline{X}_0^{(p)}/S_0) = F_{S_0}^* H_{\text{DR}}^2(\underline{X}_0/S_0)$ defined by the Cartesian square (2.5.1), and the $\mathcal{O}_{S_0}$ -linear map $\tilde{F} : H_{\text{DR}}^2(\underline{X}_0^{(p)}/S_0) \rightarrow H_{\text{DR}}^2(\underline{X}_0/S_0)$ defined by the relative Frobenius $F_{\underline{X}_0/S_0}$. It is immediate that

^{*}This is a formal variant of [4, 8.26], from which it follows easily.

$$F_{\text{Hdg}}^1 H_{\text{DR}}^2(\underline{X}_0/S_0) = H_{\text{DR}}^2(\underline{X}_0/S_0) \cap F_{\text{Hdg}}^1 H_{\text{DR}}^2(\underline{X}_0^{(p)}/S_0),$$

so it is enough to prove

$$F_{\text{Hdg}}^1 H_{\text{DR}}^2(\underline{X}_0^{(p)}/S_0) = \ker(\tilde{F} : H_{\text{DR}}^2(\underline{X}_0^{(p)}/S_0) \rightarrow H_{\text{DR}}^2(\underline{X}_0/S_0)). \quad (2.6.2)$$

By 2.2, the Hodge spectral sequence of $\underline{X}_0^{(p)}/S_0$ degenerates at E_1 , and therefore there is an exact sequence

$$0 \longrightarrow F_{\text{Hdg}}^1 H_{\text{DR}}^2(\underline{X}_0^{(p)}/S_0) \longrightarrow H_{\text{DR}}^2(\underline{X}_0^{(p)}/S_0) \xrightarrow{\pi} H^2(\underline{X}_0^{(p)}, \mathcal{O}) \longrightarrow 0, \quad (2.6.3)$$

where π is the canonical projection. Since $F_{\underline{X}_0/S_0}$ vanishes on $\Omega_{\underline{X}_0^{(p)}/S_0}^i$ for $i \geq 1$, there is a commutative square

$$\begin{array}{ccc} H_{\text{DR}}^2(\underline{X}_0^{(p)}/S_0) & \xrightarrow{\pi} & H^2(\underline{X}_0^{(p)}, \mathcal{O}) \\ \tilde{F} \downarrow & & \downarrow \tilde{F} \\ H_{\text{DR}}^2(\underline{X}_0/S_0) & \xleftarrow{(*)} & H^2(\underline{X}_0, \mathcal{H}^0(\Omega_{\underline{X}_0/S_0}^\bullet)) \end{array}$$

where $(*)$ is induced by the inclusion $\mathcal{H}^0(\Omega_{\underline{X}_0/S_0}^\bullet) \hookrightarrow \Omega_{\underline{X}_0/S_0}^\bullet$. Degeneration at E_1 of the Hodge spectral sequence of $\underline{X}_0/S_0$ entails, by the standard argument using the Cartier operator, degeneration at E_2 of the conjugate spectral sequence

$$E_2^{ij} = H^i(\underline{X}_0, \mathcal{H}^j(\Omega_{\underline{X}_0/S_0}^\bullet)) \Longrightarrow H_{\text{DR}}^*(\underline{X}_0/S_0).$$

In particular, $(*)$ is injective. The right vertical arrow is an isomorphism (a special case of the Cartier isomorphism), so $\tilde{F}$ and π have the same kernel. Together with (2.6.3), this proves the assertion.

REMARKS 2.7.

a) By [4, 8.26], one also has

$$F_{\text{Hdg}}^2 H_{\text{DR}}^2(\underline{X}_0/S_0) = \text{Im}(\{x \in H_{\text{DR}}^2(\underline{X}/S) \mid Fx \in p^2 H_{\text{DR}}^2(\underline{X}/S)\} \longrightarrow H_{\text{DR}}^2(\underline{X}_0/S_0)), \quad (2.7.1)$$

but we shall not need this. b) The F -crystal $H_{\text{DR}}^4(\underline{X}/S)$ is isomorphic to the “Tate crystal” $\mathcal{O}(-2)$, namely $\mathcal{O}_S$ equipped with $\nabla = d$ and $F = p^2\varphi$. In the absence, in the literature, of a trace morphism $H_{\text{DR}}^4(\underline{X}/S) \rightarrow \mathcal{O}_S$, one may nevertheless

see this as follows. First, $H_{\text{DR}}^4(\underline{X}/S)$ is the image under cup product of $F^1 H_{\text{DR}}^2(\underline{X}/S) \otimes F^1 H_{\text{DR}}^2(\underline{X}/S)$. By (2.5.4) and 2.6, it follows that $F(H_{\text{DR}}^4(\underline{X}/S)) \subset p^2 H_{\text{DR}}^4(\underline{X}/S)$. Since $H_{\text{DR}}^4(\underline{X}/S) \otimes W = H^4(X_0/W)$ is isomorphic to $W(-2)$, the crystal $H_{\text{DR}}^4(\underline{X}/S)$ has the form $U(-2)$, where U is a unit-root F -crystal; every unit-root F -crystal over S is trivial [7], proving the assertion.

2.8. The last ingredient needed in the proof of 1.6 is the crystalline Chern class of an invertible sheaf (cf. [3]). Let $\text{Fil}^1 \Omega_{\underline{X}/S}^\bullet$ be the subcomplex of $\Omega_{\underline{X}/S}^\bullet$ which is the kernel of the canonical projection $\Omega_{\underline{X}/S}^\bullet \rightarrow \mathcal{O}_{\underline{X}_0}$, i.e.

$$\text{Fil}^1 \Omega_{\underline{X}/S}^\bullet = (p\mathcal{O}_{\underline{X}} \xrightarrow{d} \Omega_{\underline{X}/S}^1 \xrightarrow{d} \Omega_{\underline{X}/S}^2). \quad (2.8.1)$$

The defining exact sequence for Fil^1 ,

$$0 \longrightarrow \text{Fil}^1 \Omega_{\underline{X}/S}^\bullet \longrightarrow \Omega_{\underline{X}/S}^\bullet \longrightarrow \mathcal{O}_{\underline{X}_0} \longrightarrow 0, \quad (2.8.2)$$

gives an exact sequence

$$0 \longrightarrow 1 + \text{Fil}^1 \Omega_{\underline{X}/S}^\bullet \longrightarrow (\Omega_{\underline{X}/S}^\bullet)^* \longrightarrow \mathcal{O}_{\underline{X}_0}^* \longrightarrow 0, \quad (2.8.3)$$

where

$$(\Omega_{\underline{X}/S}^\bullet)^* = (\mathcal{O}_{\underline{X}}^* \xrightarrow{d \log} \Omega_{\underline{X}/S}^1 \xrightarrow{d} \Omega_{\underline{X}/S}^2)$$

is the “multiplicative” de Rham complex. By definition, the Chern-class homomorphism

$$c_1 : \text{Pic}(\underline{X}_0) = H^1(\underline{X}_0, \mathcal{O}_{\underline{X}_0}^*) \longrightarrow H^2(\underline{X}, \text{Fil}^1 \Omega_{\underline{X}/S}^\bullet) \quad (2.8.4)$$

is the composite of the connecting morphism of (2.8.3) and the map induced by the morphism of complexes

$$\log : 1 + \text{Fil}^1 \Omega_{\underline{X}/S}^\bullet \longrightarrow \text{Fil}^1 \Omega_{\underline{X}/S}^\bullet, \quad (2.8.5)$$

which is the identity in degrees ≥ 1 and sends $1 + x$ to $\sum (-1)^i x^{i+1} / (i+1)$ in degree zero. The legitimacy of (2.8.5) is immediate.

Since $H^1(\underline{X}_0, \mathcal{O}) = 0$ (2.2(a)), the canonical map

$$H^2(\underline{X}, \mathrm{Fil}^1 \Omega_{\underline{X}/S}^\bullet) \longrightarrow H_{\mathrm{DR}}^2(\underline{X}/S) \quad (2.8.6)$$

is injective. We shall therefore sometimes regard c_1 in (2.8.4) as taking values in $H_{\mathrm{DR}}^2(\underline{X}/S)$.

PROPOSITION 2.9. Let $\underline{L}_0 \in \mathrm{Pic}(\underline{X}_0)$ and put $x = c_1(\underline{L}_0)$. Then

- a) $Fx = px$;
- b) $\nabla x = 0$.

For these formulas it is convenient to interpret c_1 crystallinely. The construction of [3, 2.1] gives a homomorphism

$$c_{1/W} : \mathrm{Pic}(\underline{X}_0) \longrightarrow H_{\mathrm{cris}}^2(\underline{X}_0, J_{\underline{X}_0/W}), \quad (2.9.1)$$

where $J_{\underline{X}_0/W}$ is the crystalline sheaf defined by the canonical exact sequence

$$0 \longrightarrow J_{\underline{X}_0/W} \longrightarrow \mathcal{O}_{\underline{X}_0/W} \longrightarrow \mathcal{O}_{\underline{X}_0} \longrightarrow 0. \quad (2.9.2)$$

By (2.3.2),

$$H^2(\underline{X}, \mathrm{Fil}^1 \Omega_{\underline{X}/S}^\bullet) = R^2(f_0)_{\mathrm{cris}*}(J_{\underline{X}_0/W})(S),$$

and hence

$$H_{\mathrm{cris}}^0(S_0, R^2(f_0)_{\mathrm{cris}*} J_{\underline{X}_0/W}) = \{x \in H^2(\underline{X}, \mathrm{Fil}^1 \Omega_{\underline{X}/S}^\bullet) \mid \nabla x = 0\}. \quad (2.9.3)$$

It is readily checked that (2.8.4) is the composite of (2.9.1) and the canonical homomorphism

$$H_{\mathrm{cris}}^2(\underline{X}_0, J_{\underline{X}_0/W}) \longrightarrow H_{\mathrm{cris}}^0(S_0, R^2(f_0)_{\mathrm{cris}*} J_{\underline{X}_0/W}), \quad (2.9.4)$$

under the identification (2.9.3). This proves 2.9(b). The preceding description also shows functoriality in $\underline{X}_0/S_0$ of (2.8.4), which implies 2.9(a), because, for the absolute Frobenius $F_{\underline{X}_0}$ of $\underline{X}_0$,

$$Fx = c_1(F_{\underline{X}_0}^* \underline{L}_0) = c_1(\underline{L}_0^{\otimes p}) = px.$$

REMARK 2.9.5. As regards (b), a reader reluctant to use crystalline considerations may verify it “by hand” from the explicit Katz–Oda description [9] of the Gauss–Manin connection. Let D be a derivation of S/W . Choose an affine open covering $\mathcal{U}$ of $\underline{X}_0$, a cocycle (g_{ij}) with values in $\mathcal{O}_{\underline{X}_0}^*$ representing $\underline{L}_0$, and lifts D_i of D to derivations of $\underline{X}/W$ on U_i . Lifting g_{ij} to a section $\tilde{g}_{ij}$ of $\mathcal{O}_{\underline{X}}^*$ on $U_i \cap U_j$ defines a 2-cocycle h_{ijk} with values in $1 + p\mathcal{O}_{\underline{X}}$; the class $x = c_1(\underline{L}_0)$ is represented by

$$(\log h_{ijk}) + (d \log g_{ij}) \in Z^2(\mathcal{U}, \mathrm{Fil}^1 \Omega_{\underline{X}/S}^\bullet).$$

On the other hand, by [9], $\nabla(D)x$ is the class of the 2-cocycle $(i < j < k)$

$$y = (D_i \log h_{ijk}) + (dD_i \log g_{ij}) - ((D_i \log - D_j \log)g_{jk}).$$

One checks that y is the coboundary of the 1-cochain $(D_i \log \tilde{g}_{ij}) \in C^1(\mathcal{U}, \Omega_{\underline{X}/S}^1)$, and therefore $\nabla(D)x = 0$.

2.10. Functoriality of the crystalline Chern class implies that, for every W -valued point e of S , there is a commutative square

$$\begin{array}{ccc} \mathrm{Pic}(\underline{X}_0) & \xrightarrow{c_1} & H_{\mathrm{DR}}^2(\underline{X}/S) = H_{\mathrm{cris}}^2(\underline{X}_0/S) \\ \downarrow \scriptstyle \underline{L} \mapsto \underline{L}|_{X_0} & & \downarrow \scriptstyle e^* \\ \mathrm{Pic}(X_0) & \xrightarrow{c_1} & H_{\mathrm{DR}}^2(X/W) = H_{\mathrm{cris}}^2(X_0/W), \end{array} \quad (2.10.1)$$

where X/W is obtained from $\underline{X}/S$ by base change along e . Since X_0 is a K3 surface, $\mathrm{Pic}(X_0)$ coincides with the Néron–Severi group $\mathrm{NS}(X_0)$ and is torsion free; hence the lower horizontal arrow in (2.10.1) is injective (3.4).

2.11. Proof of 1.6. Suppose that L_0 extends to an invertible sheaf $\underline{L}_0$ on $\underline{X}_0$, and let $x = c_1(\underline{L}_0) \in H_{\text{DR}}^2(\underline{X}/S)$. Commutativity of (2.10.1) gives $e^*x = c_1(L_0)$. Since L_0 is nontrivial by hypothesis, $c_1(L_0) \neq 0$ by the preceding remark, and hence $x \neq 0$. By 2.9, $Fx = px$ and $\nabla x = 0$. Let p^n be the highest power of p dividing x , and put $y = p^{-n}x$; the image y_0 of y in $H_{\text{DR}}^2(\underline{X}_0/S_0) = H_{\text{DR}}^2(\underline{X}/S) \bmod p$ is nonzero. We still have $Fy = py$ and $\nabla y = 0$. By (2.6.1), the first relation implies $y_0 \in F_{\text{Hdg}}^1 H_{\text{DR}}^2(\underline{X}_0/S_0)$. Since $y_0 \neq 0$ and $\nabla y_0 = 0$, one cannot have $y_0 \in F_{\text{Hdg}}^2 H_{\text{DR}}^2(\underline{X}_0/S_0)$, for this would contradict the injectivity of

$$\text{gr}^2 \nabla : F_{\text{Hdg}}^2 H_{\text{DR}}^2(\underline{X}_0/S_0) \longrightarrow \Omega_{S_0/k}^1 \otimes \text{gr}_F^1 H_{\text{DR}}^2(\underline{X}_0/S_0)$$

from 2.4.4. Thus the image of y_0 in $\text{gr}_F^1 H_{\text{DR}}^2(\underline{X}_0/S_0)$ is nonzero; but $\nabla y_0 = 0$, contradicting the fact that

$$\text{gr}^1 \nabla : \text{gr}_F^1 H_{\text{DR}}^2(\underline{X}_0/S_0) \longrightarrow \Omega_{S_0/k}^1 \otimes \text{gr}_F^0 H_{\text{DR}}^2(\underline{X}_0/S_0)$$

is an isomorphism. This contradiction proves 1.6.

3. APPENDIX; CRYSTALLINE CHERN CLASSES AND INTERSECTIONS*

The purpose of this appendix is to establish that, on a proper smooth surface over k , the intersection number of two divisors is the trace of the cup product of their crystalline Chern classes (this compatibility apparently does not occur in the literature).

3.1. Let X be a proper smooth scheme over k . Recall [2, VI 3.3.6] that to every closed subscheme Y of X , smooth over k and of codimension d in X , one can associate a cohomology class

$$c\ell(Y) \in H^{2d}(X/W). \quad (3.1.1)$$

If Y' and Y'' are smooth closed subschemes of codimensions d' and d'' , meeting transversely, then [2, VI 4.3.15]

*By P. Deligne and L. Illusie.

$$c\ell(Y') c\ell(Y'') = c\ell(Y' \cap Y'') \quad (3.1.2)$$

in $H^{2(d'+d'')}(X/W)$. If X' is proper and smooth over k , $f : X' \rightarrow X$ is a k -morphism, and Y is a smooth closed subscheme of codimension d such that $Y \times_X X'$ is smooth of codimension d in X' , then [2, VI 4.3.13]

$$c\ell(Y \times_X X') = f^* c\ell(Y). \quad (3.1.3)$$

Finally, if a is a rational point of X/k , then [2, VII 3.1.7]

$$\mathrm{Tr}_{X/W}(c\ell(a)) = 1. \quad (3.1.4)$$

In particular, if X is connected, $c\ell(a)$ is independent of a , since $\mathrm{Tr}_{X/W} : H^{2n}(X/W) \rightarrow W$ is an isomorphism, where $n = \dim X$ [2, VII 2.1.1].

3.2. To every vector bundle L on X one can associate Chern classes [3, 2.4]

$$c_i(L) \in H^{2i}(X/W), \quad (3.2.1)$$

which vanish for $i > \mathrm{rk}(L)$. These classes depend functorially on L : if $f : X' \rightarrow X$ is a k -morphism, then (loc. cit.)

$$c_i(f^*L) = f^*c_i(L). \quad (3.2.2)$$

If D is a smooth effective divisor on X , there are therefore both the cohomology class $c\ell(D) \in H^2(X/W)$ and the Chern class $c_1(\mathcal{O}_X(D)) \in H^2(X/W)$.

PROPOSITION 3.2.3. Let D be a smooth effective divisor on X . If $\mathcal{O}_X(D)$ is very ample, then

$$c\ell(D) = c_1(\mathcal{O}_X(D)). \quad (3.2.3.1)$$

REMARK 3.2.4. Berthelot (unpublished) verified that (3.2.3.1) holds without the hypothesis that $\mathcal{O}_X(D)$ be very ample. The proof is rather complicated. For our purposes, 3.2.3 is sufficient.

Let us prove 3.2.3. By 3.1.3 and 3.2.2, we reduce at once to the case in which X is projective space $\mathbf{P}_k^N$ and D is a hyperplane. Put $\mathbf{P}_W^N = X'$, and let D' be a hyperplane of X' lifting D . Then $H^2(X/W) = H_{\text{DR}}^2(X'/W)$. By [2, VI 3.3.5], $c\ell(D)$ is the de Rham cohomology class of D' , which is computed from the extension of $\Omega_{D'/W}^\bullet[-1]$ by $\Omega_{X'/W}^\bullet$ supplied by the de Rham complex of X'/W with logarithmic poles along D' . If $(t_0, \dots, t_N)$ are homogeneous coordinates on X' , it follows readily that $c\ell(D)$ is represented, on the standard covering, by the 1-cocycle

$$\left(\frac{dt_j}{t_j} - \frac{dt_i}{t_i} \right)_{i < j}.$$

On the other hand, by [3, 2.3], the Chern class $c_1(\mathcal{O}_{X'}(1)) \in H_{\text{DR}}^2(X'/W)$ is obtained from the cocycle t_j/t_i by the homomorphism $d \log : \mathcal{O}_{X'}^* \rightarrow \Omega_{X'/W}^\bullet[1]$. Thus $c_1(\mathcal{O}_X(D)) = c\ell(D)$.

THEOREM 3.3. Let X be a proper smooth surface over k . If D_1, D_2 are divisors on X , then

$$(D_1 \cdot D_2) = \text{Tr}_{X/W}(c_1(\mathcal{O}_X(D_1)) c_1(\mathcal{O}_X(D_2))),$$

where $(D_1 \cdot D_2)$ denotes their intersection number.

If D_1, D_2 are effective, smooth, very ample, and transverse, the conclusion follows from (3.1.2), (3.1.4), and 3.2.3. To reduce to this case, choose a very ample divisor H on X . For n sufficiently large, $D_1 + nH$ and $D_2 + nH$ are very ample. By Bertini, there are effective smooth very ample divisors D'_1, D''_1, D'_2, D''_2 such that

$$\mathcal{O}_X(D_1) = \mathcal{O}_X(D'_1 - D''_1), \quad \mathcal{O}_X(D_2) = \mathcal{O}_X(D'_2 - D''_2),$$

and one may further suppose that the four pairs

$$(D'_1, D'_2), \quad (D'_1, D''_2), \quad (D''_1, D'_2), \quad (D''_1, D''_2)$$

are transverse. Additivity of c_1 and bilinearity of the intersection pairing reduce the theorem to the special case already considered, completing the proof.

3.4. Again let X be a proper smooth scheme over k . It is not known whether, for two smooth closed subschemes of X of the same codimension, algebraic equivalence implies equality of their crystalline cohomology classes (this is known only for zero-dimensional subschemes; cf. 3.1.4!). On the other hand, if $\mathrm{NS}(X) = \mathrm{Pic}(X)/\mathrm{Pic}^0(X)$ is the Néron–Severi group, the homomorphism

$$c_1 : \mathrm{Pic}(X) \longrightarrow H^2(X/W)$$

vanishes on $\mathrm{Pic}^0(X)$, because $\mathrm{Pic}^0(X)$ is p -divisible and $H^2(X/W)$ is of finite type over W . It therefore induces

$$c_1 : \mathrm{NS}(X) \longrightarrow H^2(X/W). \quad (3.4.1)$$

Suppose now that X is a surface. By the Hodge index theorem, the intersection form on $\mathrm{NS}(X) \otimes \mathbf{Q}$ is nondegenerate. In view of 3.3, it follows that

$$c_1 \otimes \mathbf{Q} : \mathrm{NS}(X) \otimes \mathbf{Q} \longrightarrow H^2(X/W) \otimes \mathbf{Q} \quad (3.4.2)$$

is injective.

REMARK 3.5. One can show [6, II 5.10] that (3.4.1) defines an injection $\mathrm{NS}(X) \otimes \mathbf{Z}_p \hookrightarrow H^2(X/W)$. It is probably not true in general that (3.4.1) induces an injection $\mathrm{NS}(X)/p\mathrm{NS}(X) \rightarrow H^2(X/W)/pH^2(X/W)$, i.e. that the de Rham Chern-class map sends $\mathrm{NS}(X)/p\mathrm{NS}(X)$ injectively into $H_{\mathrm{DR}}^2(X/k)$. This does hold if $H^*(X/W)$ is torsion free and the Hodge-to-de Rham spectral sequence of X/k degenerates at E_1 ; see [13, 1.4] and [6, II 5.18]. These conditions are satisfied, for example, by a K3 surface.

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